

CENTRE FOR QUANTUM TECHNOLOGIES · NATIONAL UNIVERSITY OF
SINGAPORE

Large Language Models

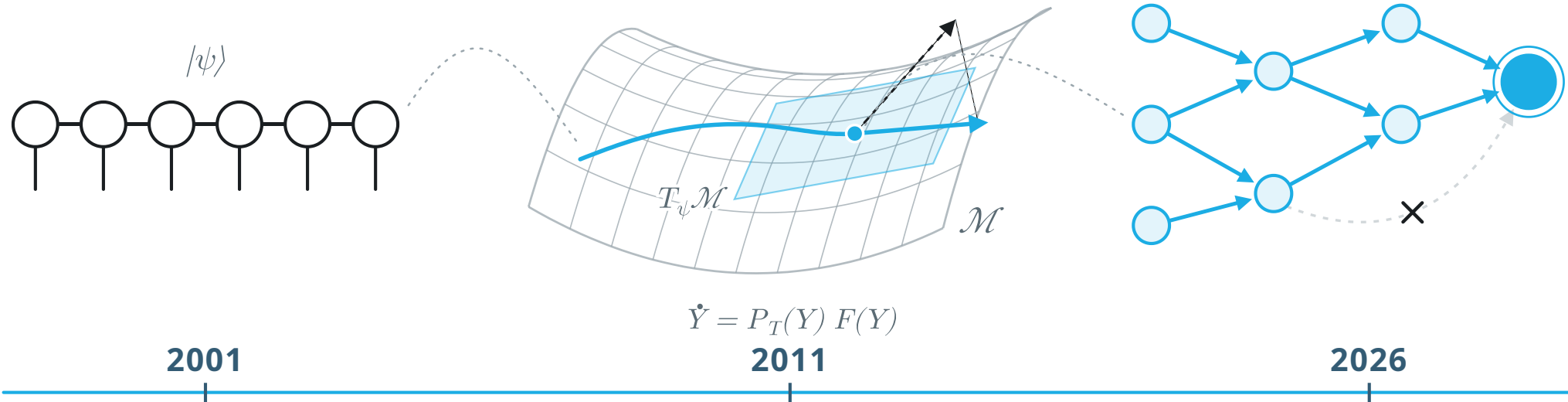
A physicist's perspective

Tobias J. Osborne · Leibniz Universität Hannover

8 October 2026 · Singapore

Quick survey: hands up

- Who has used ChatGPT, Claude, or Gemini?
 - Who has used one for research?
 - Who has used a coding agent?



Quantum information and tensor networks since **2001**; professor of theoretical physics in **Hannover since 2010**.

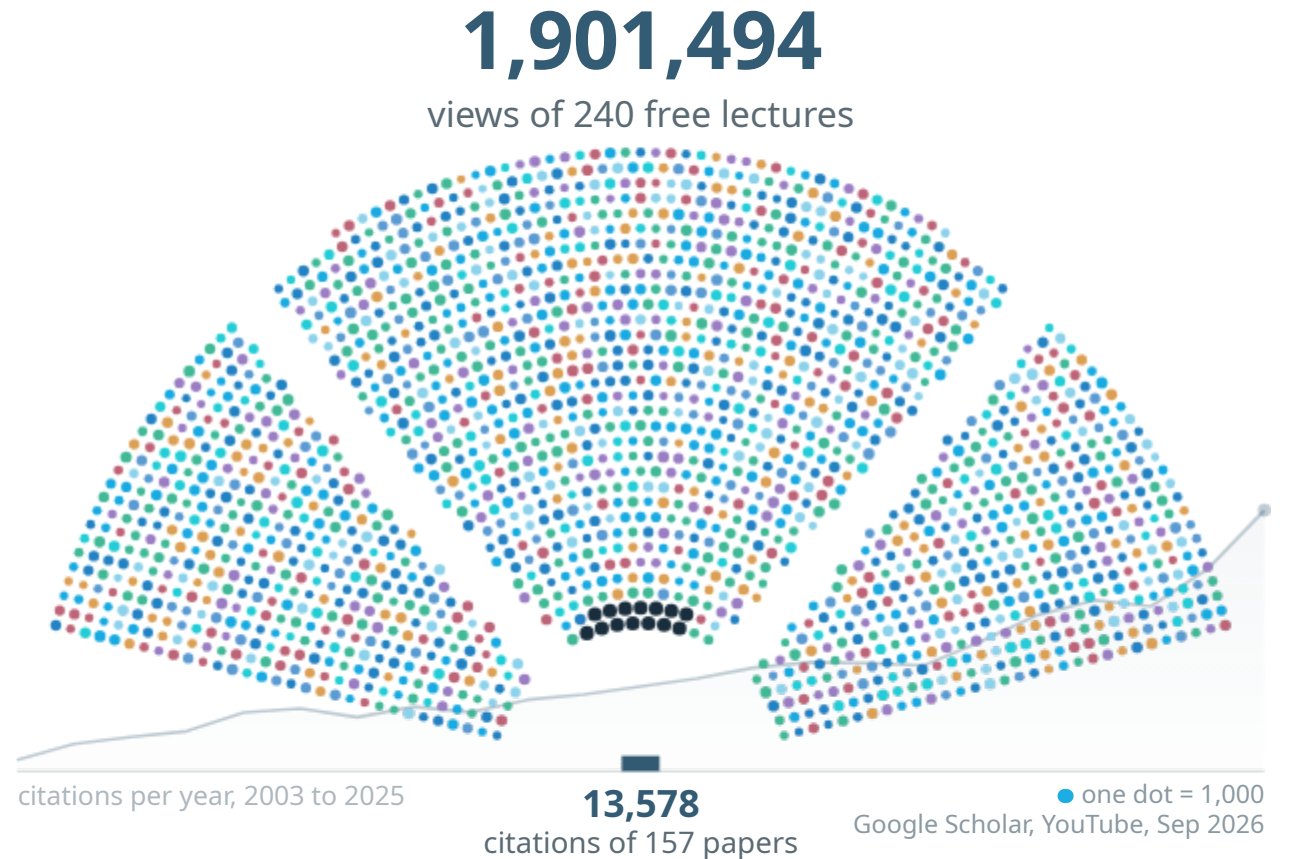
I helped develop the time-dependent variational principle for matrix product states.

Now I am **actively engaging with LLMs and science**.

Open lectures

My lectures are **free to everyone**, and they have been watched far more often than my papers have been cited.

YouTube youtube.com/@tobiasjosborne
Blog tjoresearchnotes.wordpress.com



PART ONE

Why this talk?

It started as a drop

February 13, 2026 Research Publication

GPT-5.2 derives a new result in theoretical physics

In a new preprint, GPT-5.2 proposed a formula for a gluon amplitude later proved by an internal OpenAI model and verified by the authors.

[Read the preprint ↗](#)



Listen to article | 3:54

[Share](#)

We've published a new preprint showing that a type of particle interaction many physicists expected would not occur can in fact arise under specific conditions. The work focuses on gluons, the particles that carry the strong nuclear force. The [preprint](#) is available on arXiv and is being submitted for publication. In the meantime, we welcome feedback from the community.

arXiv:2602.12176 · Harvard, Vanderbilt, Cambridge, OpenAI · February 2026
gauge-theory scattering amplitudes: GPT-5.2 conjectured the closed form and proved it

[Submitted on 20 May 2026]

Remarks on the disproof of the unit distance conjecture

Noga Alon, Thomas F. Bloom, W. T. Gowers, Daniel Litt, Will Sawin, Arul Shankar, Jacob Tsimerman, Victor Wang, Melanie Matchett Wood

We present a short, digested, human-verified version of the recent OpenAI-generated counterexample to the Erdős unit distance conjecture, and a sequence of reflections on it. The argument relies crucially on ideas that may, at least in retrospect, be attributed to Ellenberg-Venkatesh, Golod-Shafarevich, and Hajir-Maire-Ramakrishna.

Comments: 19 pages; expository note based on very recent developments

Subjects: **Combinatorics (math.CO)**; Number Theory (math.NT)

Cite as: [arXiv:2605.20695](#) [[math.CO](#)]

(or [arXiv:2605.20695v1](#) [[math.CO](#)] for this version)

<https://doi.org/10.48550/arXiv.2605.20695> 

Submission history

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Access Paper:

[View PDF](#)[HTML \(experimental\)](#)[TeX Source](#) [view license](#)

Current browse context:

math.CO

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Change to browse by:

[math](#)

[math.NT](#)

References & Citations

[NASA ADS](#)

[Google Scholar](#)

[Semantic Scholar](#)

Then, a trickle

machine-generated results in 2026, one dot each

● Lean artefact ○ no Lean ○ physics and neighbours

Generation has been automated
faster than verification.

T3 openai/math: 722 manuscripts, one repo 17 fields

6 Oct · 372 result families · 26.0M lines of Lean · unreleased model
Lean for 235 families, mostly a fragment · **zero referees**

T1* Navier–Stokes blowup with smooth forcing PDE

Sep · a Millennium Problem, in Lean · OpenAI, 10,000 agents, unreleased model
12 hours after Buckmaster and Alpöge's Euler blowup: a priority fight

T1 Zeta zeros: two thirds on the line, up from 41.6%

Aug · "Proof discovered autonomously by Claude" · Lean · reproved by hand, 2 Sep

T1* "Astra": ten decade-open problems

Aug · OpenAI · Lean, under \$2k each · **zero referees**

T1 Erdős #728

Jan · first autonomous solve, with a Lean certificate

T2 HAWK-256 broken crypto

Jul · Mythos Preview, 2^{64} to 2^{38} ; withdrawn from NIST within a day

T3 Complex structure on S^6 , open since 1947

Jul · 108 pages, self-hosted, no arXiv; reportedly written out by Claude

T2 Jacobian conjecture false in $\dim \geq 3$

Jul · "hello there the jacobian conjecture is false thanx": the entire announcement, on X

T2+ Jamming exponents, a 12-year-old identity physics

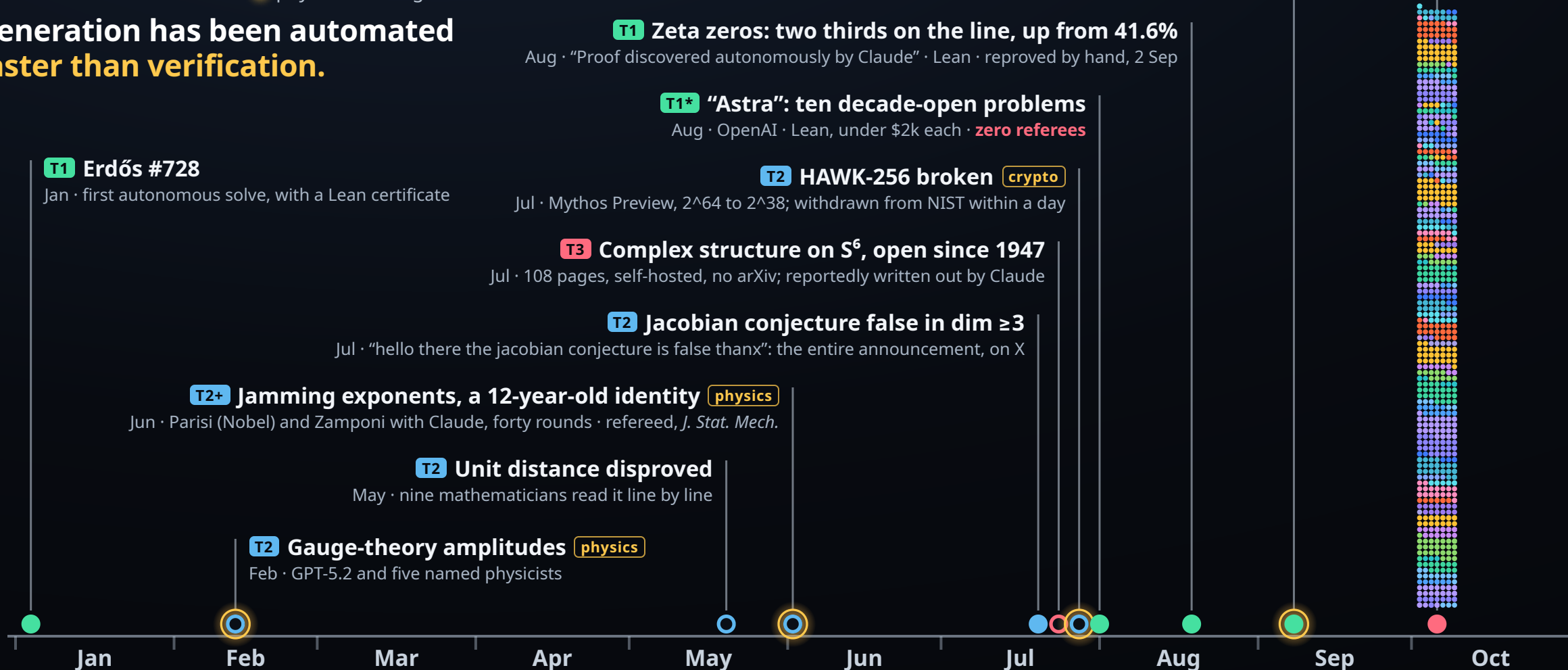
Jun · Parisi (Nobel) and Zamponi with Claude, forty rounds · refereed, *J. Stat. Mech.*

T2 Unit distance disproved

May · nine mathematicians read it line by line

T2 Gauge-theory amplitudes physics

Feb · GPT-5.2 and five named physicists



T1 machine-checked or officially graded **T1*** Lean, no referee yet **T2** named humans read it **T2+** refereed journal **T3** announced only

dates of first public announcement · sources: arXiv, *J. Stat. Mech.*, lab and press announcements, github.com/openai/math

The trickle became a flood

github.com/openai/math · one dot per manuscript, by folder date

722
manuscripts

372
result families

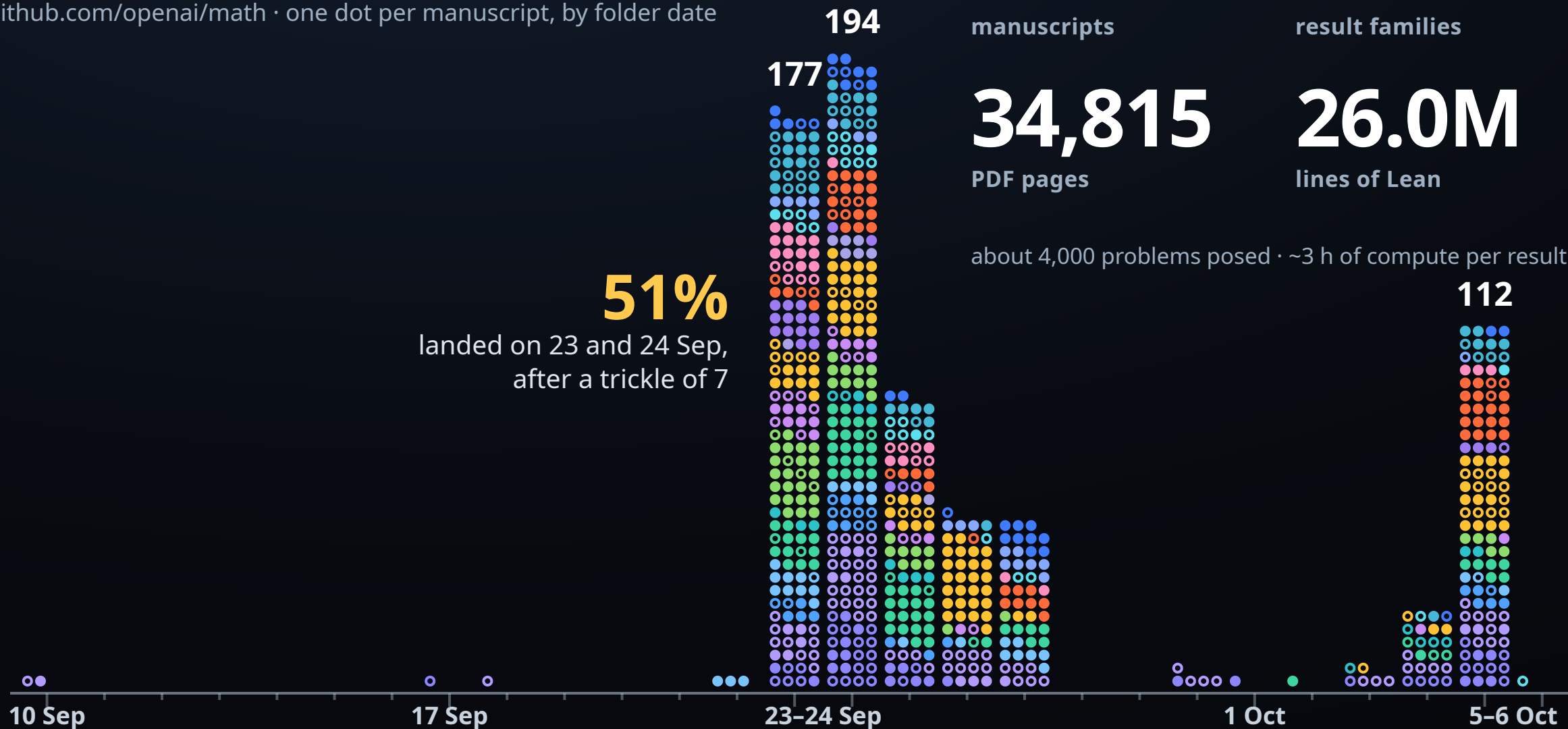
34,815
PDF pages

26.0M
lines of Lean

about 4,000 problems posed · ~3 h of compute per result

51%

landed on 23 and 24 Sep,
after a trickle of 7

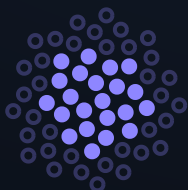


● family has a Lean artefact ○ no Lean ■ physics blocks ■ everything else

github.com/openai/math, cloned 7 Oct 2026 · dates from preprint folder names · totals from the repo README and a file scan

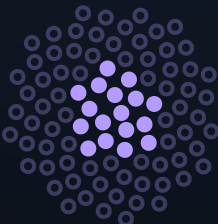
17 disciplines

% of families with a Lean artefact. Often a fragment, not the headline.
MUB(6) = 3: the Lean covers only $N(6) \leq 5$.



Number theory

52% with Lean



Algebraic & complex geometry

19% with Lean



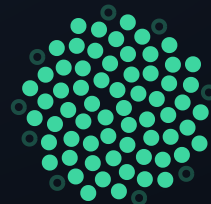
Real & complex analysis

56% with Lean



Convex & metric geometry

87% with Lean



Theoretical CS

80% with Lean



Dynamics & ergodic

75% with Lean



Combinatorics

89% with Lean



Algebra

50% with Lean



Math logic

100% with Lean



Group theory

86% with Lean



Topology

17% with Lean



Functional analysis

91% with Lean



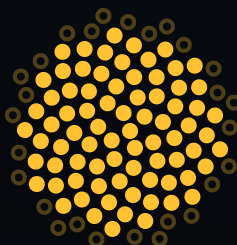
Differential geometry

52% with Lean



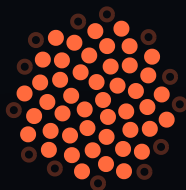
PDE

69% with Lean



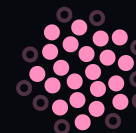
Probability & stat mech

66% with Lean



Math physics

68% with Lean



Operator algebras

74% with Lean

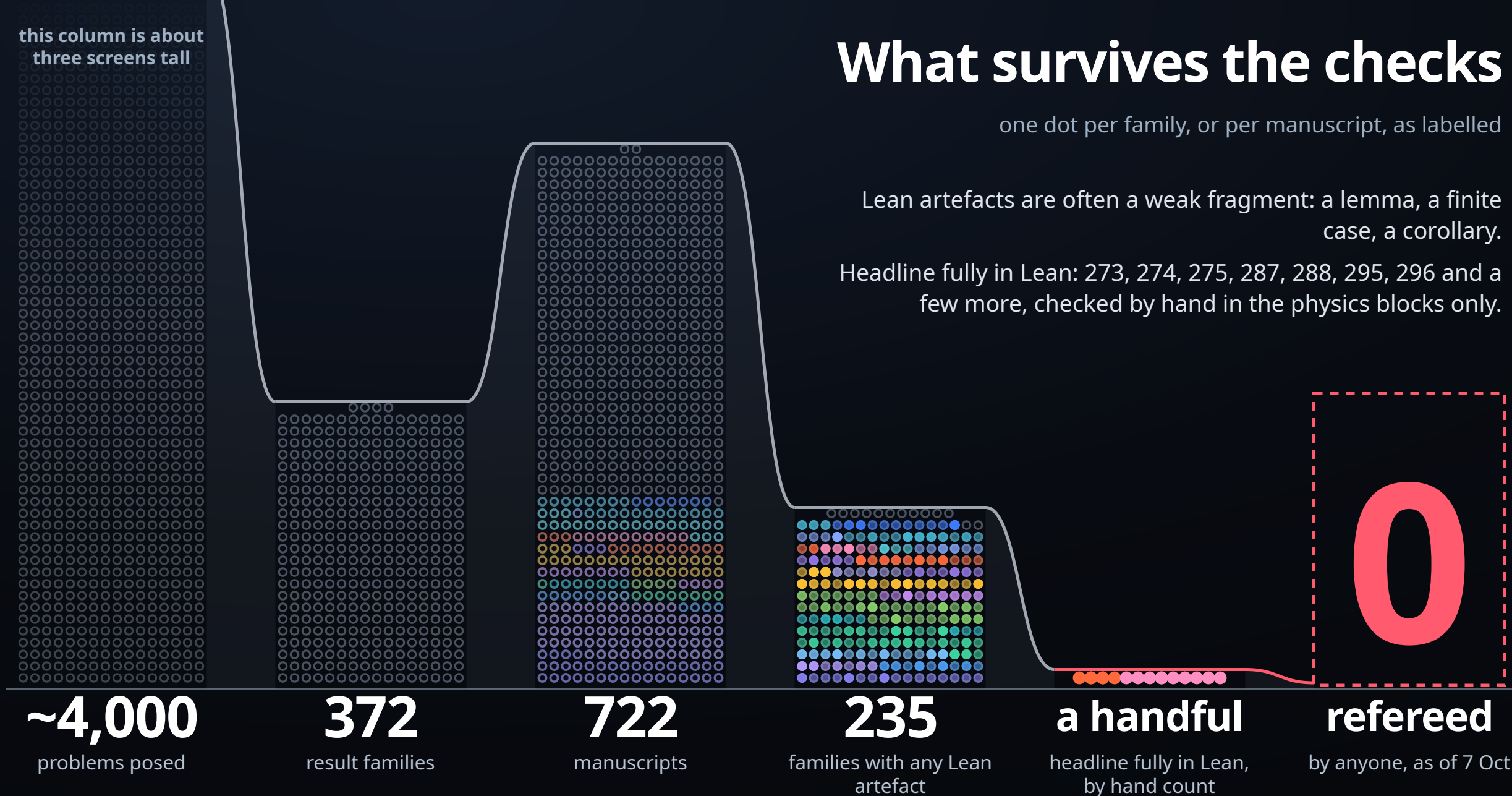
this column is about
three screens tall

What survives the checks

one dot per family, or per manuscript, as labelled

Lean artefacts are often a weak fragment: a lemma, a finite case, a corollary.

Headline fully in Lean: 273, 274, 275, 287, 288, 295, 296 and a few more, checked by hand in the physics blocks only.



The physicist's picks

8 of 372 families · theorem statements read from the LaTeX proofs skimmed, not checked

FAMILY 265

2D area law

$S(A) \leq C |\partial A|$ on any finite piece of $\mathbb{Z}^2$, from a global gap alone.

LEAN: §2 ONLY

FAMILY 268

COMPUTER-ASSISTED

Spin-1 Haldane gap

Unique ground state and a uniform gap for even periodic chains, $L \geq 60$.

NO LEAN

FAMILY 271

Heisenberg ferromagnet

$d \geq 3$, any spin: order at low T , and Bloch's $T^{3/2}$ law exactly.

LEAN: ORDER ONLY

FAMILY 269

COMPUTER-ASSISTED

Laughlin $\nu = 1/3$ gap

Uniform gap on the sphere, stable under projected disorder.

LEAN: NO DISORDER

FAMILY 266

COMPUTER-ASSISTED

MUBs in dimension 6

$N(6) = 3$. The exclusion of four bases is a binary64 computation.

LEAN: $N(6) \leq 5$ ONLY

FAMILY 273

Entropy photon-number inequality

The Guha-Erkmen-Shapiro conjecture for the beam splitter.

LEAN: HEADLINE

FAMILY 274

Parity $\notin \text{QAC}^0$

Moore's conjecture, measured-output model. Two proofs.

LEAN: HEADLINE

FAMILY 275

QMA-hard Coulomb energy

Full continuum molecular Hamiltonian, even with unit charges.

LEAN: BOTH THEOREMS

HEADLINE

FRAGMENT

NONE

what the Lean covers

3 of 8 have the headline in Lean. The rest need a referee.

Deep Blue comes to mathematics

a name for a feeling,
borrowed from Simon Willison

Deep Blue *n.*

the ennui, sliding into dread, of watching a machine do what you spent years learning to do
coined on Oxide and Friends (credit Adam Leventhal) · written up by Simon Willison, about software engineers

“What am I even for?”

MAY 1997, NEW YORK

IBM's Deep Blue beats the world champion, Garry Kasparov, 3½ to 2½.

Calculation, the skill that defined mastery, stops being scarce.

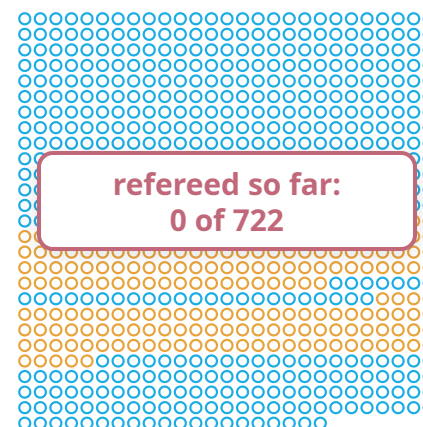
Afterwards: the engine became every player's coach. People still play people.

CHESS 1997



game 6, 11 May 1997:
Kasparov resigns after 19 moves

2026 MATHEMATICS



one dot per manuscript · **amber:**
the three physics blocks, 195

6 OCT 2026, OPENAI/MATH

722 manuscripts, 372 families,
17 disciplines, 26.0M lines of Lean.

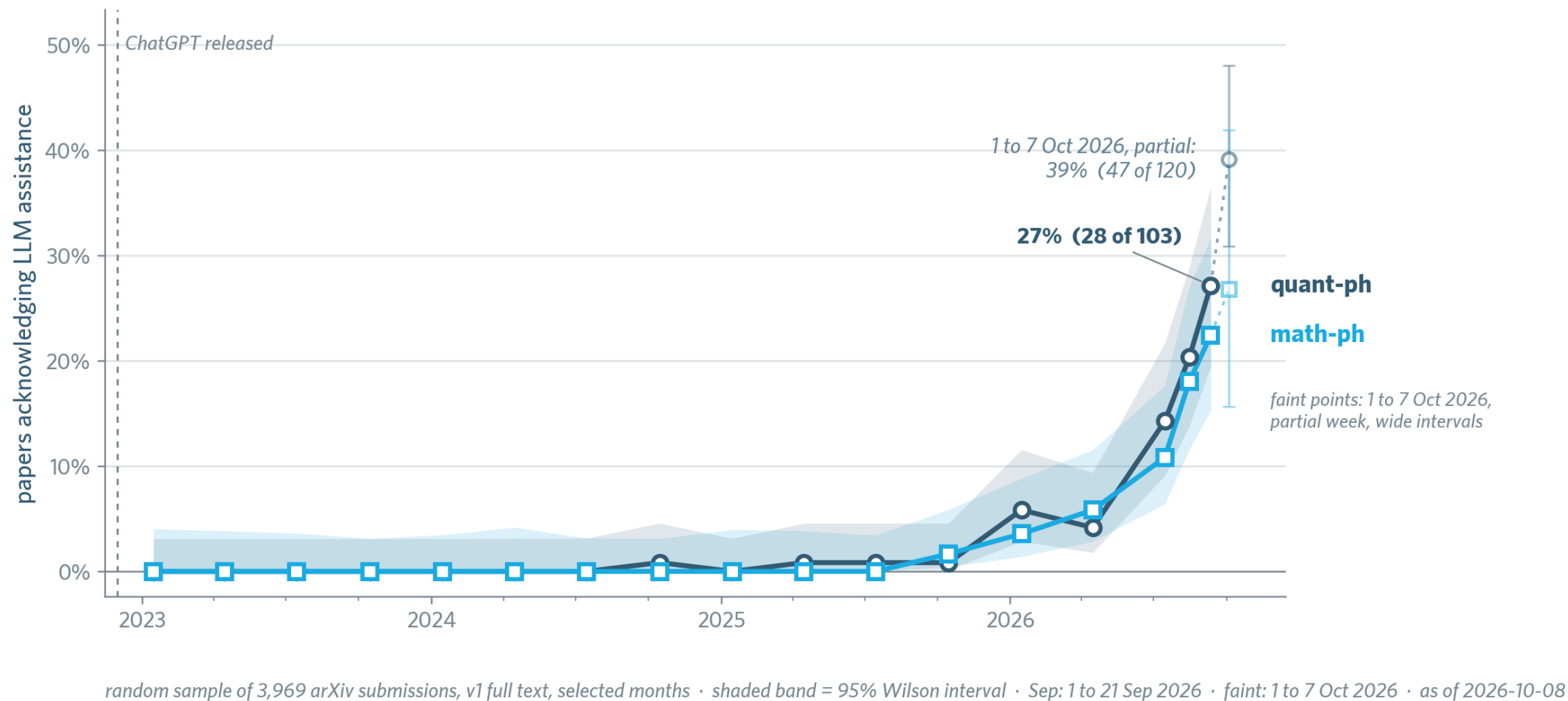
Proving stops being scarce. In the pile: the spin-one Haldane gap, the 2D area law, parity $\notin$ QAC⁰.

Afterwards: being decided this week.

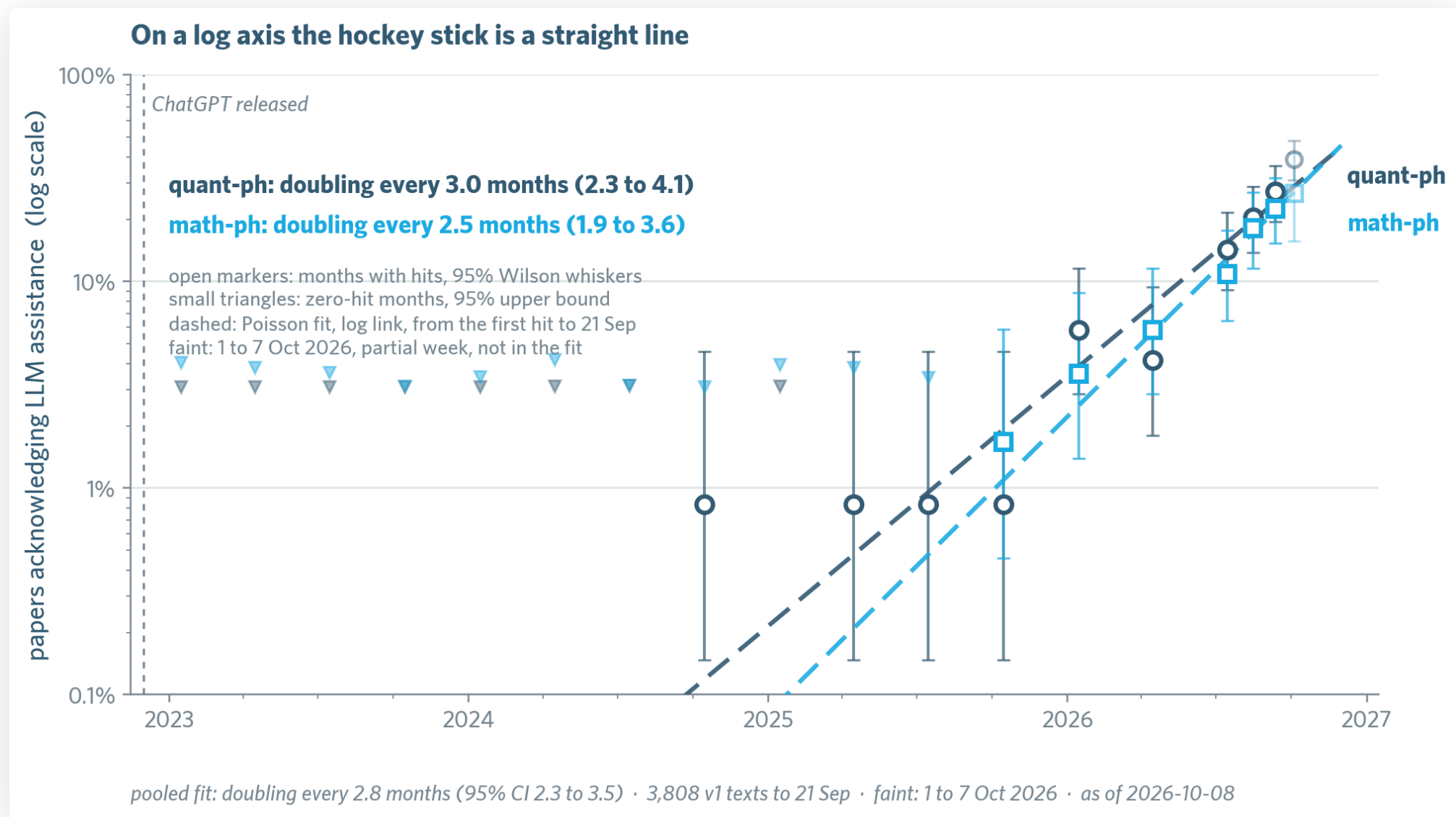
Where the mirror breaks: chess had a scoreboard. Mathematics has a referee queue.

“All of the chess players and the Go players went through this a decade ago and they have come out stronger.” Simon Willison

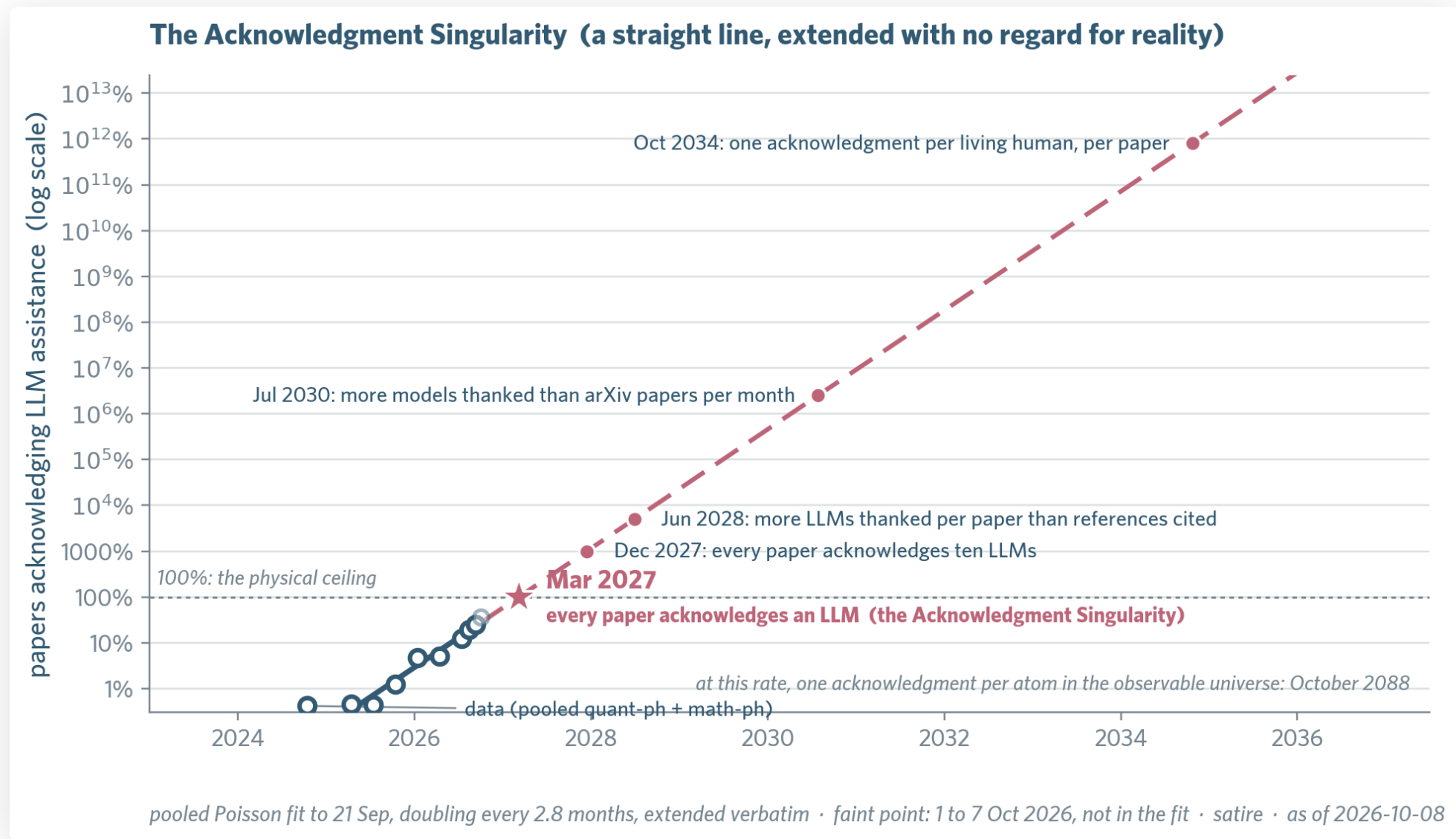
In chess the engine became the coach. That is the use of agents I want to show you.



Papers whose acknowledgments name an LLM: quant-ph **0% through 2023, 27% in September 2026**, 39% in the first week of October
 3,969 sampled v1 submissions, every hit read by hand · a disclosure rate, not a usage rate · faint: 1 to 7 Oct (math-ph only 41 papers) · 8 Oct 2026



Same data, log axis: the rate has **doubled every 2.8 months** since the first hit (95% CI 2.3 to 3.5)
fit to 21 Sep, then tested on 1 to 7 Oct: the line predicted 28%, the week came in at 36% (95% CI 29 to 44) · no bend yet · 8 Oct 2026



Extend the line and every paper thanks an LLM by **March 2027**; by late 2034 each paper thanks one model per living human
honest reading: a disclosure rate hits its ceiling within months, so the line must bend. Early October landed above it · satire, 8 Oct 2026

A control experiment, run on us

First-year theoretical physics, six years of grades

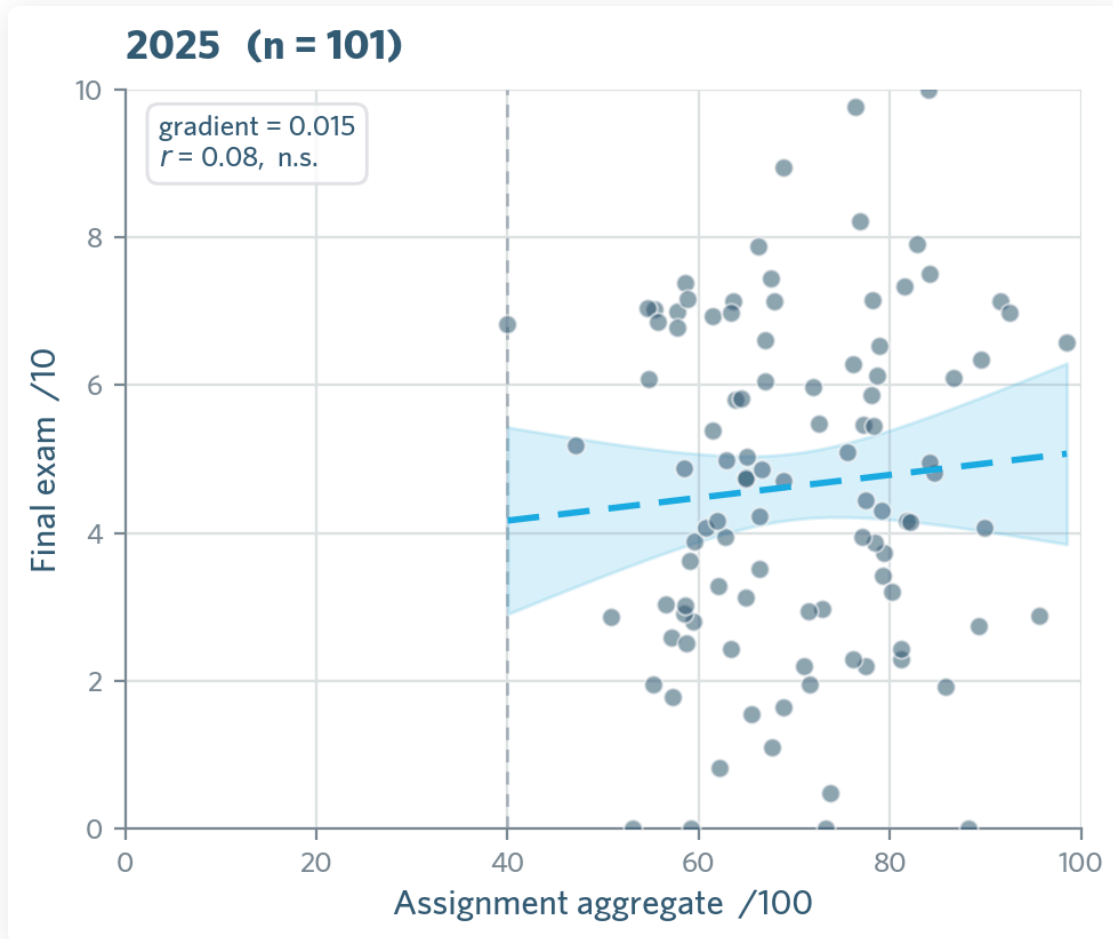
Weekly take-home assignment, out of 100 (unsupervised)

Invigilated final exam, out of 10 (the control)

2020 to 2023: the two agree at **$b \approx 0.10$**

Synthetic data, illustrative: reproduces the structure of the real six-year analysis.

Six years, one control experiment



2025

GRADIENT B

0.015

CORRELATION

R

0.08

$r = 0.08$, not significant: the assignment no longer measures what the exam measures.

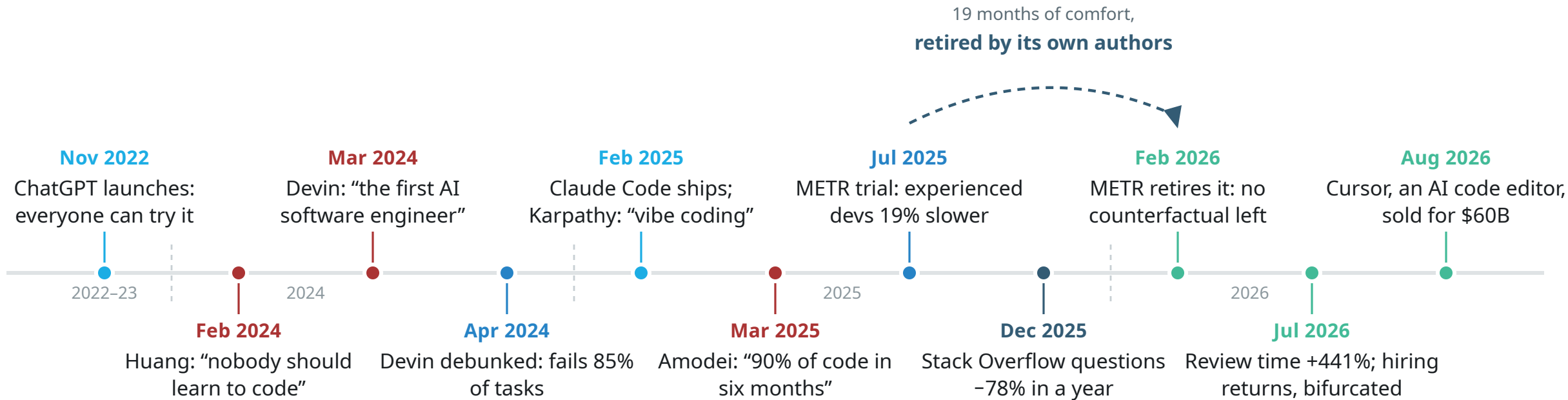


press → or drag to step through the years

Synthetic data, illustrative: reproduces the structure of the real six-year analysis.

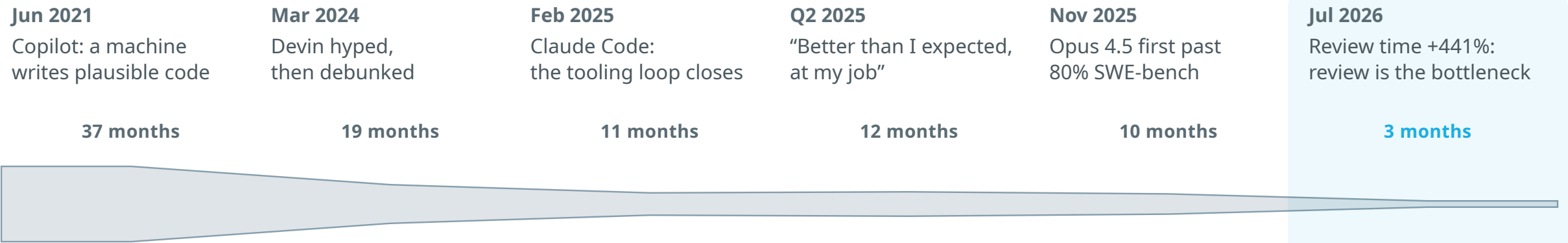
The timeline

software engineering
Nov 2022 → Aug 2026

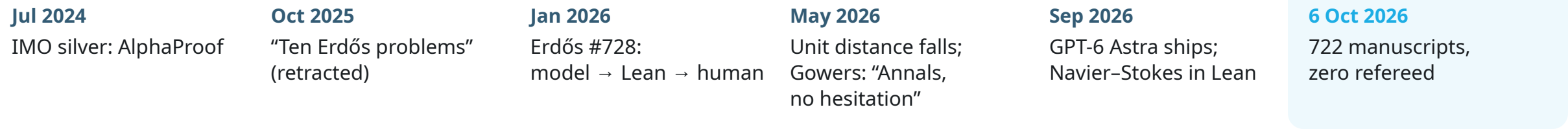


The lag collapsed: from three years to three months

SOFTWARE ENGINEERING



MATHEMATICS & THEORETICAL PHYSICS



You are here

8 October 2026, Singapore

Offset ~3 years in 2024 · ~10 months in September · ~3 months now

Both fields now wait on the same thing: a human to check. In quant-ph, 27% of September papers thank an LLM.

**Progress is astonishing.
Let us understand them.**

PART TWO

What is an LLM?

**A new kind of resource:
general-purpose intelligence,
bought by the token**

(kind of)

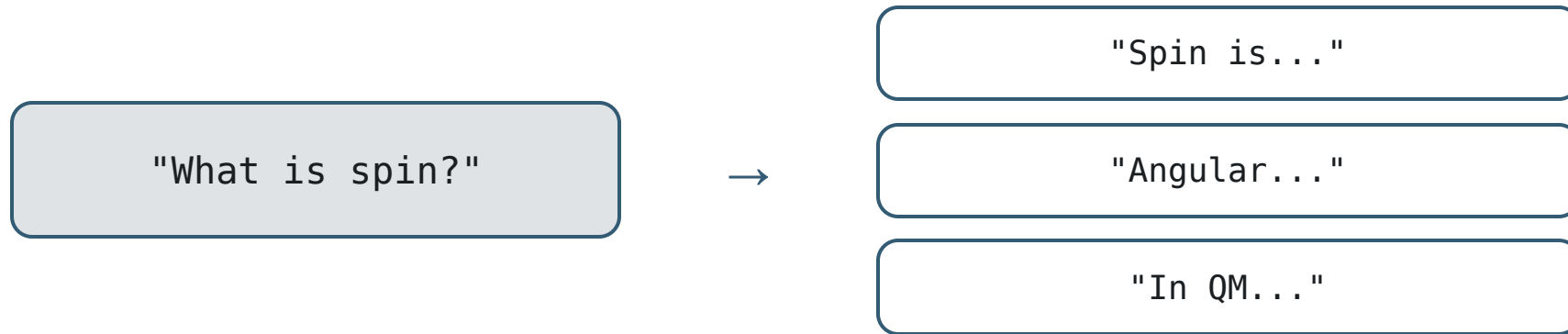
From outside, looking in

```
f : String -> String
```

Outside the API boundary, this is the observable behaviour.

No hidden state is carried between calls.

Nondeterministic



Same input, different outputs: the model samples from **$P(\text{next token} \mid \text{context})$**

Temperature

$$P(\text{token}_{\textcolor{red}{i}}) = \exp(\ell_{\textcolor{red}{i}}/T) / \sum_{\textcolor{red}{j}} \exp(\ell_{\textcolor{red}{j}}/T)$$

spin		0%
charge		0%
mass		0%
momentum		0%
energy		0%
a		0%
color		0%
flavour		0%

Temperature 1.0



Sample

Autoplay

The electron has

Direct analogy to the Boltzmann distribution. Low T freezes into the ground state; high T melts toward uniform.

Tokens

Not characters. Not words.

Subword chunks from a fixed vocabulary **V** of about 100k entries.

```
'theoretical physics' -> ['theor', 'etical', ' physics']
```

More precisely: **f : Token* → Token***

An LLM is a stateless, nondeterministic function

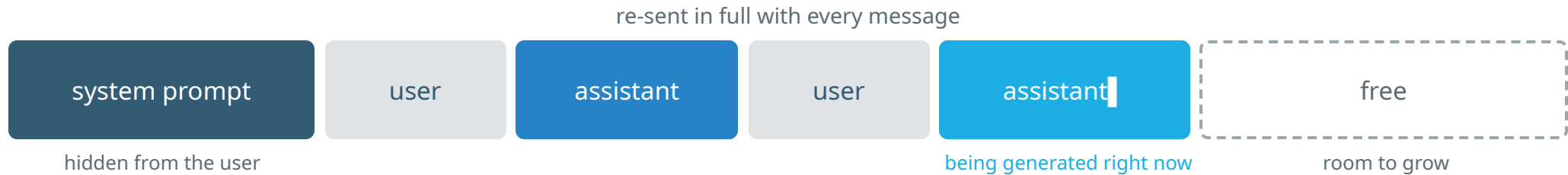
f : String -> String

that samples from $P(\text{next_token} \mid \text{context})$

via a single forward pass through a frozen neural network

The context window

Everything the model sees is **one token sequence**, capped at a fixed length: the **context window**



Current models: **200k to 1M tokens** (a million tokens is roughly two thousand pages)

Whoever controls the context controls what the model knows

PART THREE

The illusion of chat

A single API call

```
curl https://api.anthropic.com/v1/messages \
  -H "x-api-key: $KEY" \
  -H "content-type: application/json" \
  -H "anthropic-version: 2023-06-01" \
  -d '{"model": "claude-sonnet-5", "max_tokens": 256,
      "messages": [{"role": "user", "content": "What is a quark?"}]}'
```

One string in, one string out: the entire interaction

How chat works

Turn 1: [user_1]

Turn 2: [user_1, asst_1, user_2]

Turn 3: [user_1, asst_1, user_2, asst_2, user_3]

The client re-sends the **entire** history every turn: the LLM has no memory.

INTERLUDE

Failure modes

3½

The failure zoo: architectural

Hallucination

No way to guarantee the fidelity of information stored in the weights, and what comes out is *sampled*, not looked up.

Nonzero error rate

Every token is drawn from a probability distribution. Prose tolerates that; deduction and code do not.

Training cutoff

The weights are frozen once training ends. No continual learning: recall of anything later is guesswork.

Context rot

Attention degrades as the window fills. In our experience quality starts to sag beyond roughly **30%** usage.

The failure zoo: post-training

Hyperfixation (reward hacking)

Rewarded for visible completion: it deletes the failing test, hard-codes the answer, declares success.

Sycophancy

Raters prefer being agreed with. So it preferentially asserts the falsehoods *you would quite like to be true*.

Confident wrongness

Fluent, specific, real-shaped, false, and unhedged.
Never ask a model to check its own output.

Semantic shift

It imports assumptions from a better-studied neighbour: Jordan algebras silently treated as C^* -algebras.

These are architectural facts

Each one admits a structural countermeasure.

There is a way to correct for these errors

Reliable computation out of unreliable components.

PART FOUR

From function to agent

The agent loop

```
while not done:

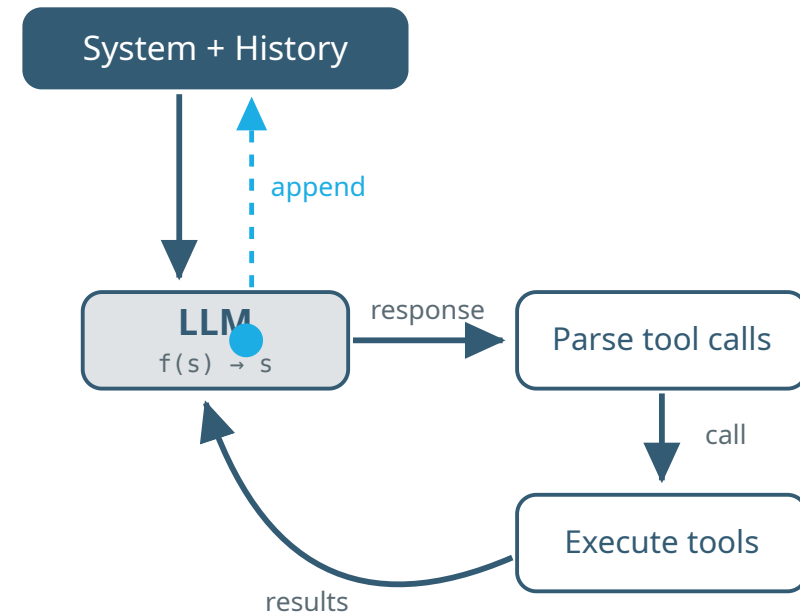
    response = llm(system_prompt + history)

    tool_calls = parse(response)

    results = execute(tool_calls)

    history += [response, results]
```

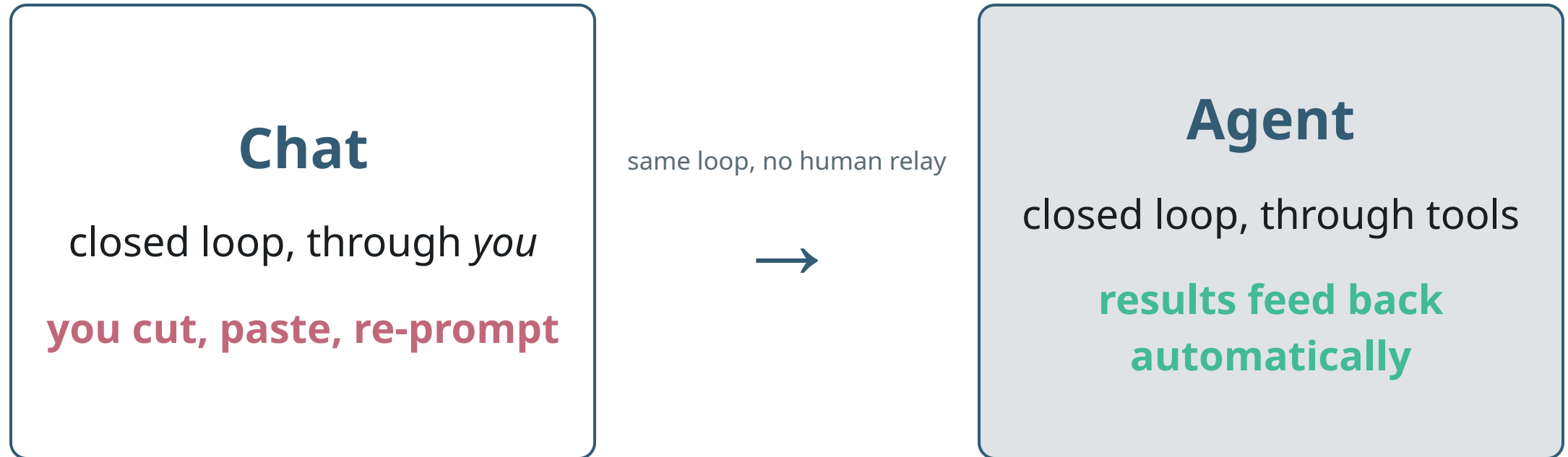
The complete architecture of every coding agent



The LLM never executes anything

It generates text; your scaffolding acts on it.

Closing the loop



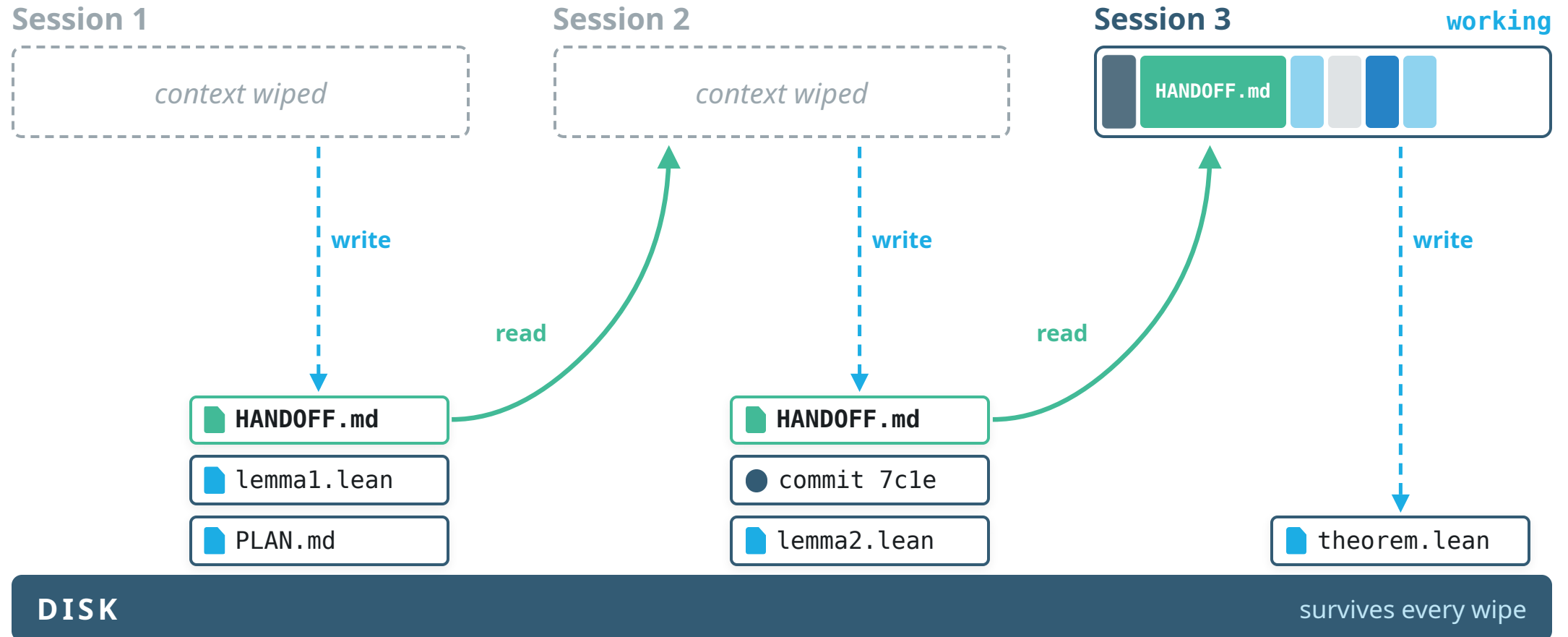
The agent closes the same loop faster, with no human cutting and pasting

Error correction for LLMs

FAILURE MODE	COUNTERMEASURE
Finite context · no continual learning	→ filesystem as memory
Hallucination	→ ground truth on disk (every cited paper kept locally)
Nonzero error rates	→ recursive decomposition and verification
Hyperfixation · sycophancy	→ independent adversarial verification (in a fresh session)
Confident wrongness	→ structured proofs (Lamport)
Semantic shift	→ formalisation
Lack of taste	→ ask a human

Every cure on the right is a tool the agent loop already gives us

The filesystem is permanent memory



Every session forgets. The files do not.

Errors compound

A flat chain of thought has **no error detection**. If each step succeeds with probability $1 - \epsilon$:

$$P(\text{the whole derivation}) = (1 - \epsilon)^n$$

PER-STEP ERROR

1%

STEPS

100

SURVIVE INTACT

37%

A slip at step 3 propagates silently to step 40, while the trace stays locally fluent throughout.

This is why long unchecked derivations rot

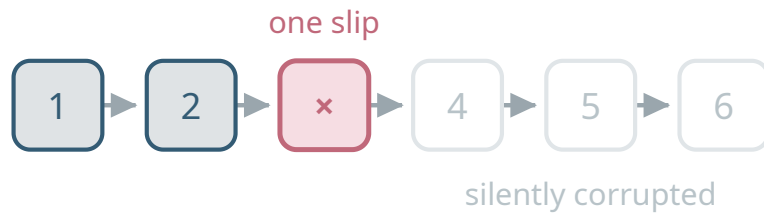
Subagents



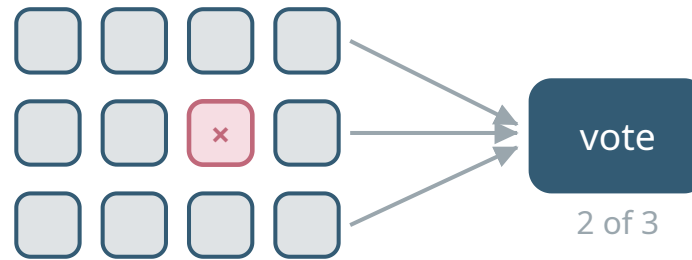
Each worker gets a clean window; results return as short reports.

Context isolation is the point

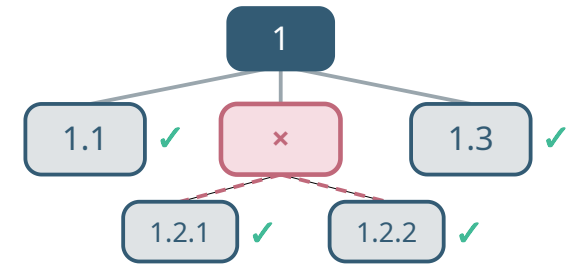
Three ways to spend compute on reliability



(a) flat chain. One slip at step 3; everything after it is quietly wrong.



(b) parallel repetition. Majority vote over final answers. Helps, but scales poorly with depth.



(c) recursive decomposition. Verify each block in isolation; expand the one that fails.

(c) is the concatenated-code strategy (von Neumann, 1956).

Above the verification threshold, reliability *compounds* instead of decaying

Why structure a proof?

Most proofs in physics and mathematics are written as flowing prose:

“By linearity ... and therefore ... which contradicts ... hence the result follows.”

This works well enough when the proof is short. But even a five-line argument can hide a gap, one that only surfaces when someone pushes back on a specific step.

And then the reader is left asking: *which* step was wrong?

Lamport structured proofs

```
1      the claim
      1.1      ASSUME ...
      1.2      ...                                [1.1]
            1.2.1      ...
      1.3      QED                                [1.2]
```

Every assertion gets a **hierarchical number**: 1, 1.1, 1.1.1 ...

Every step is justified: by citing earlier steps, or by its own children.

ASSUME / **CASE** / **QED** open and close scopes.

A leaf with no justification is an **exposed gap**.

Auditable

"I don't believe 1.3.2"

Modular

sub-steps refine independently

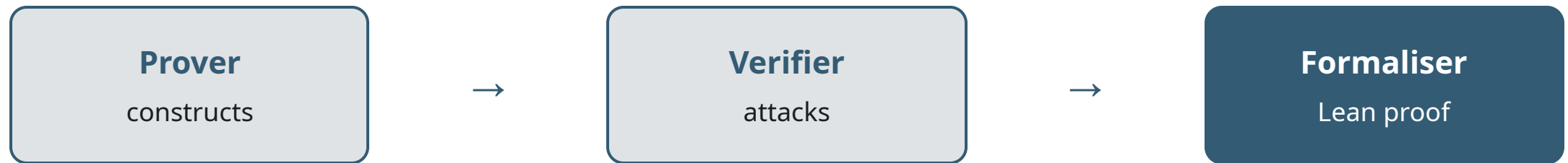
Gap-visible

nowhere for a hole to hide

That is the entire formalism: no logic syntax to learn

*L. Lamport, "How to Write a 21st Century Proof", J. Fixed Point Theory Appl. **11**, 43–63 (2012)*

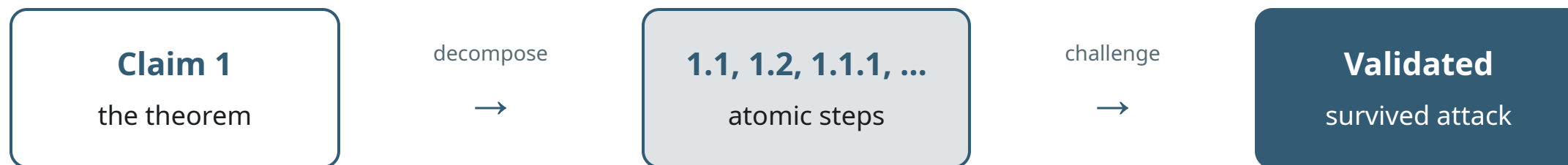
Adversarial verification



Natural language is an inadequate verification mechanism

Formal proof = a compiler for mathematics

Vibefeld: molecular pedantry



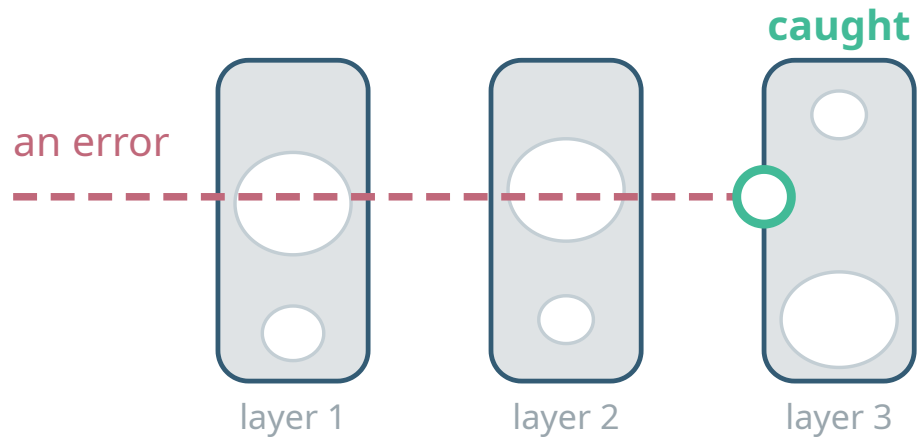
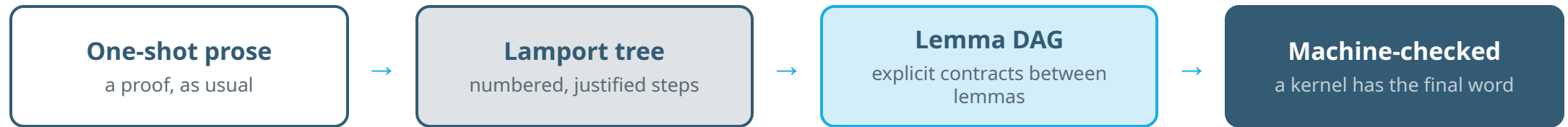
A natural-language proof, decomposed into a ledger of tiny numbered claims; each must survive adversarial challenge *on its own*.

A hallucinated proof has nowhere to hide

github.com/tobiasosborne/vibefeld

Defense in depth

Formalisation is not a single act. It is a spectrum of artifacts, increasing in cost and in rigour:

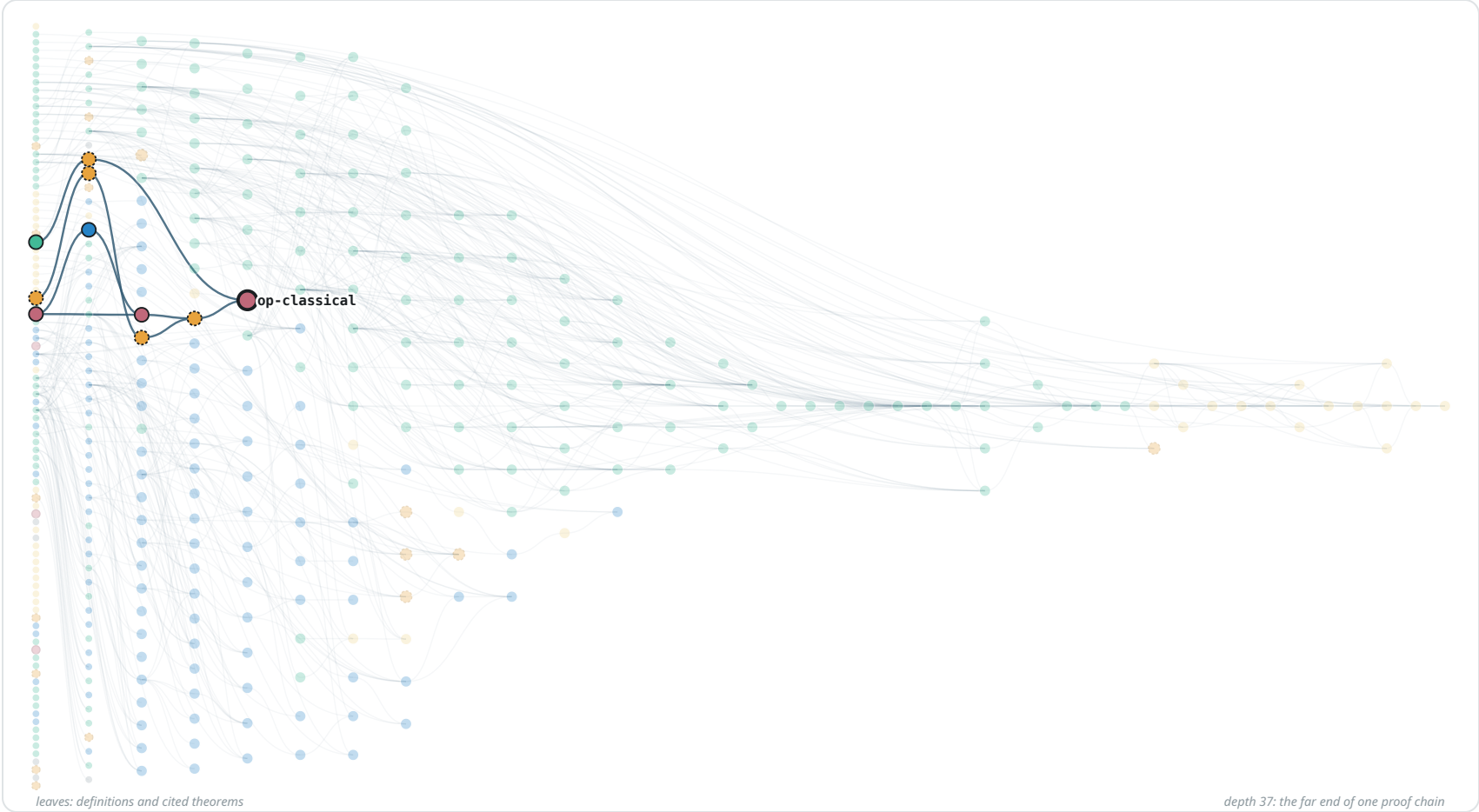


No single layer is trustworthy alone. But the layers fail in **different places**: compose them. Swiss cheese (Reason, 2000).

An agent produces every layer almost for free

What the Swiss cheese produces

one open problem, worked by agents for five weeks
364 results · 769 dependency edges · every arrow is a checked contract



click any node: what it rests on lights up · depth grows to the right

● machine-validated: proved by one fresh agent, attacked by another	169
● proved in prose, not yet adversarially checked	113
● inherited paper-proof, unaudited here	22
● conjecture, stated, heuristic or numerical	48
● open problem	6
● disproved or obstruction	6

op-classical

OPEN-PROBLEM

OPEN

(OPEN) Classical projection stability: there are universal $\eta_0, C > 0$ (n -free) such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty} \leq \eta \leq \eta_0$ admits a stochast...

rests on 9 results: 1 machine-validated · 1 prose · 7 unverified or open

The open problem itself. What it still rests on is orange and red: the gap is visible, named, and exactly where the next agent gets sent.

One agent proves, a fresh one attacks, a linker keeps the ledger. The colour is what survived.

Agents are amazing at helping you understand

One of the 722: family 268, a proof that the spin-1 Heisenberg ring has a Haldane gap

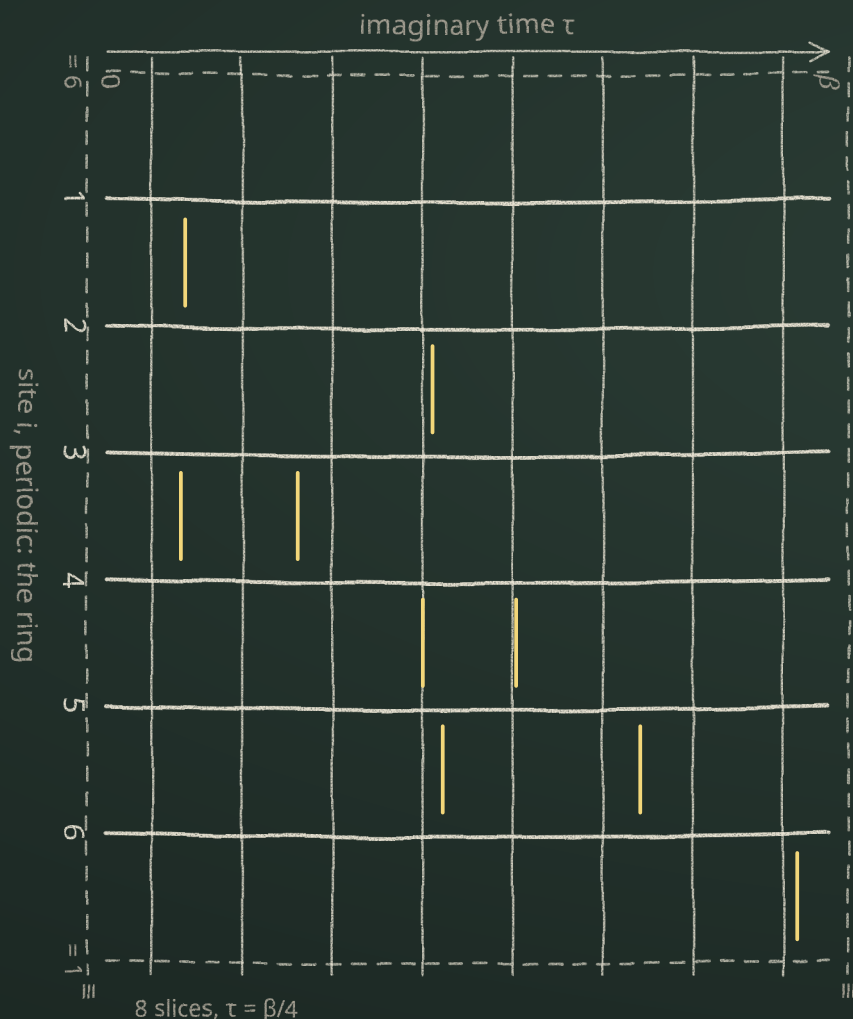
READ	One agent refereed it from the LaTeX in a single pass and re-ran the paper's exact-arithmetic checks.
BUILD	Three agents, in parallel, each turned the paper, the referee report, an explainer and a one-page brief into a five-slide interactive explainer.
TIME	All three were back within 40 minutes, yesterday evening. I picked one.

Not a recording: the toys on the next five slides compute **live, in this browser**.

One partition function, two readings

$$Z_n(\beta) = \text{Tr } e^{-\beta H_n}, H_n = \sum S_i S_{i+1} \text{ on a ring}$$

toy: spin-½ ring, $n = 6$, $\beta = 2$ (keys [] change β , M changes the slices)



rotated: the operator now reads across bonds, one column at a time

Six worldlines, one per site. Height β . Bond events arrive as a Poisson process, rate 1 per bond per unit β : $e^{-\beta H} = e^{-\beta n/4} e^{\beta \sum Q_b}$, expanded in time-ordered singlet projectors Q_b . (Aizenman–Nachtergaele; the ticks are an illustration, the numbers below are not.)

READ ALONG TIME

Gibbs weights over 64 energy levels: $Z = \sum_k e^{-\beta E_k} = \mathbf{657.1559026}$ (exact diagonalisation, uncut ring)

Cut β into $2M = 8$ slices (toy lattice): $Z^{\text{lat}} = \text{Tr}[(T_A T_B)^M]$, 64×64 matrices = **681.135416929577**

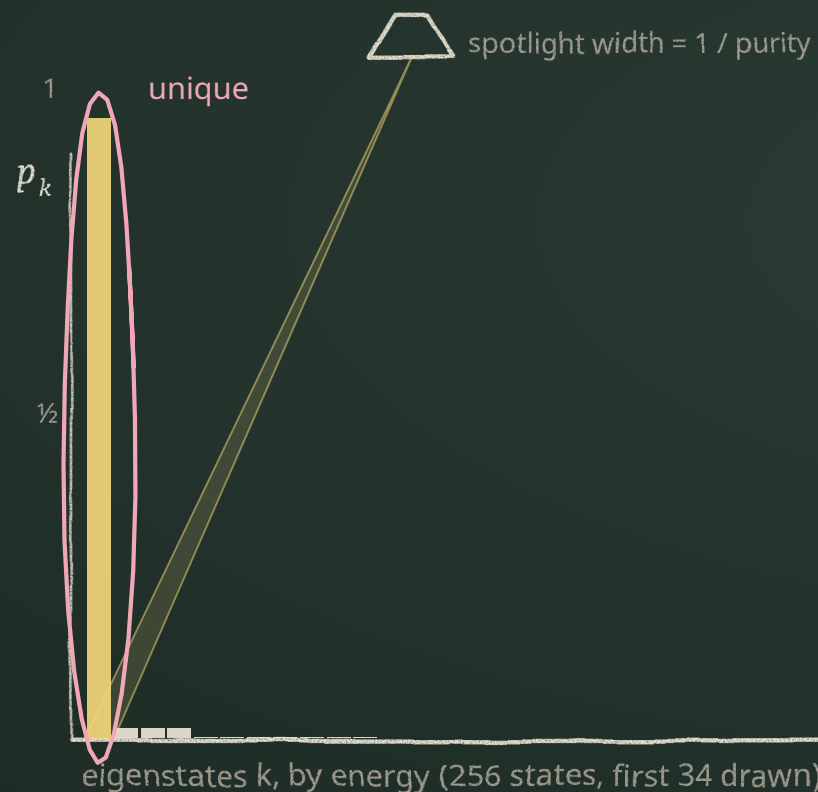
READ ALONG SPACE

Transfer across one bond, acting on a column of 8 spins (256×256): $Z^{\text{lat}} = \text{Tr}[(X_e X_o)^{n/2}] = \mathbf{681.135416929576}$

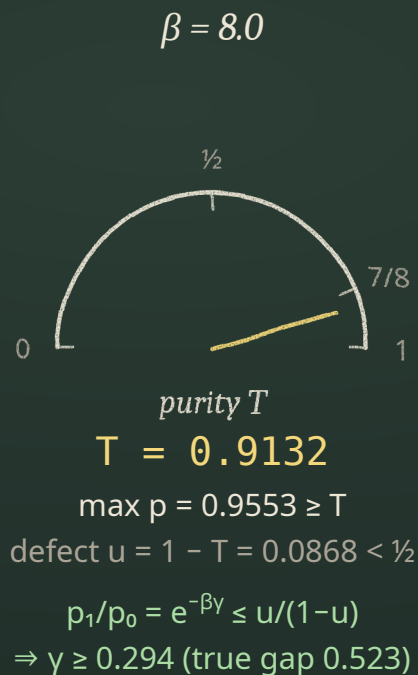
Same number to 15 digits. Finer slices, $M = 1, 2, 3, 4$: 1136.9, 758.17, 700.36, 681.14 $\rightarrow$ 657.16 (uncut).

One number, two probability vectors: $p_k \propto e^{-\beta E_k}$ over levels, $q_i \propto |\lambda_i|^n$ over the transfer spectrum.

Purity as a spotlight

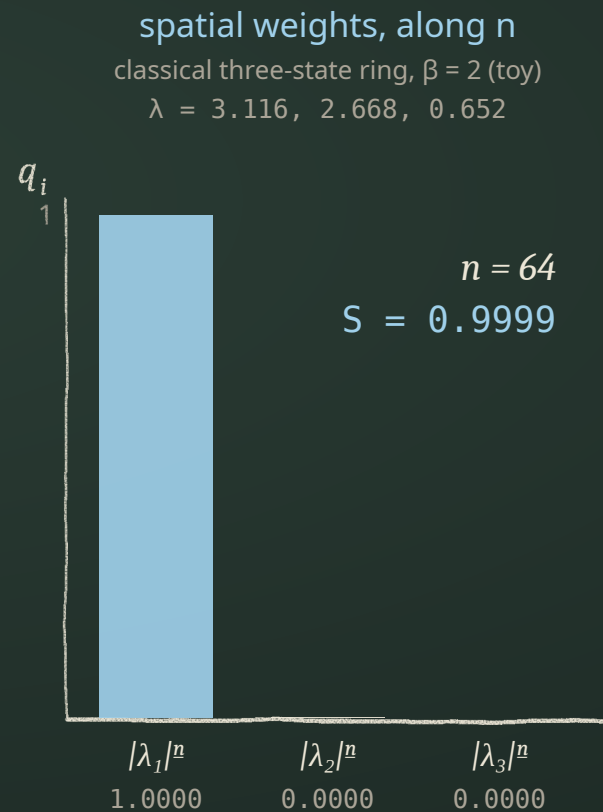


222 more: 0.000



$$\text{Gibbs purity } T(n, \beta) = \sum_k p_k^2 = Z_n(2\beta) / Z_n(\beta)^2$$

$$\text{spatial purity } S(n, \beta) = \sum_i q_i^2 = Z_{2n}(\beta) / Z_n(\beta)^2 \text{ (keys [] for } \beta, - \text{ for } n)$$

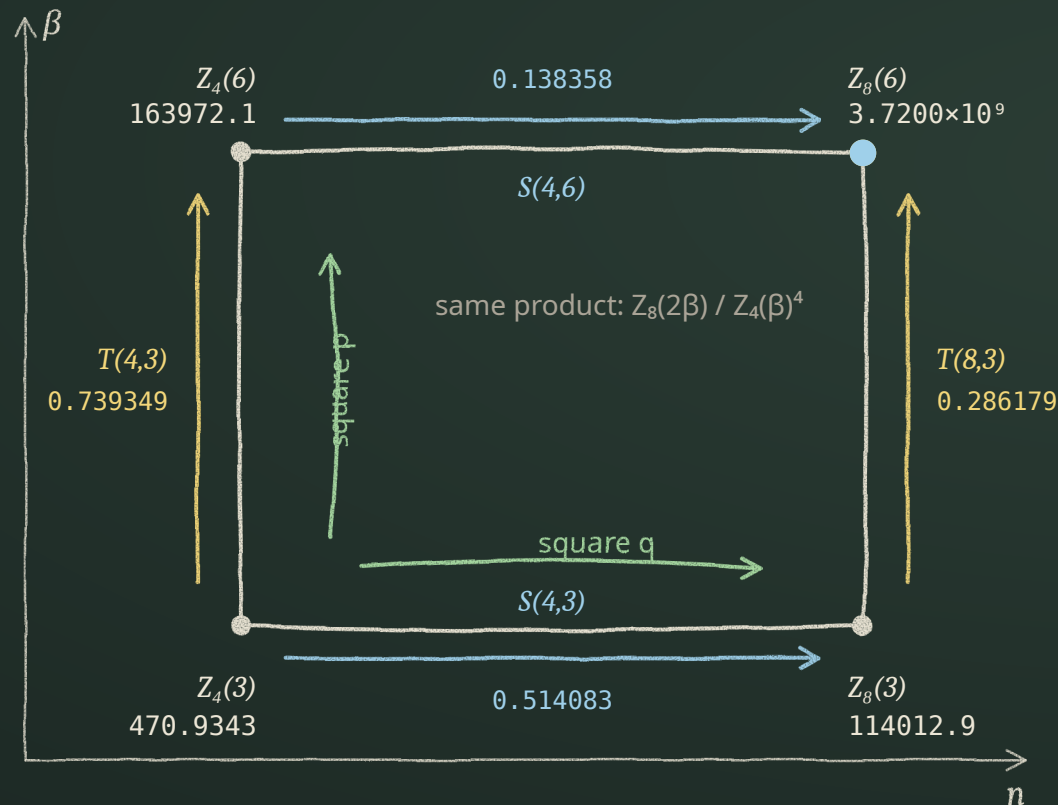


The whole problem: show $T(L, \beta)$ near 1 with β growing no faster than $\log(1/u)$.

Purity $\geq 1 - u$ with $u < 1/2$ forces one state of weight $\geq 1 - u$: a unique ground state, and $e^{-\beta\gamma} \leq u/(1-u)$.

The square on the blackboard

four partition functions, four purities, one identity
toy: spin-½ rings of $n = 4$ and $2n = 8$ sites, $\beta = 3$ (keys [] change β)



THE CANCELLATION

$$S(n, 2\beta) T(n, \beta)^2 = \frac{Z_{2n}(2\beta)}{\cancel{Z_n(2\beta)^2}} \cdot \frac{\cancel{Z_n(2\beta)^2}}{Z_n(\beta)^4} = \frac{Z_{2n}(2\beta)}{Z_n(\beta)^4}$$

$$S(n, \beta)^2 T(2n, \beta) = \frac{\cancel{Z_{2n}(\beta)^2}}{Z_n(\beta)^4} \cdot \frac{Z_{2n}(2\beta)}{\cancel{Z_{2n}(\beta)^2}} = \frac{Z_{2n}(2\beta)}{Z_n(\beta)^4}$$

Every Z once upstairs, once downstairs. Live: left route

0.07563170461608, right route 0.07563170461608.

SQUARING MOVES EACH PURITY ALONG ITS OWN AXIS

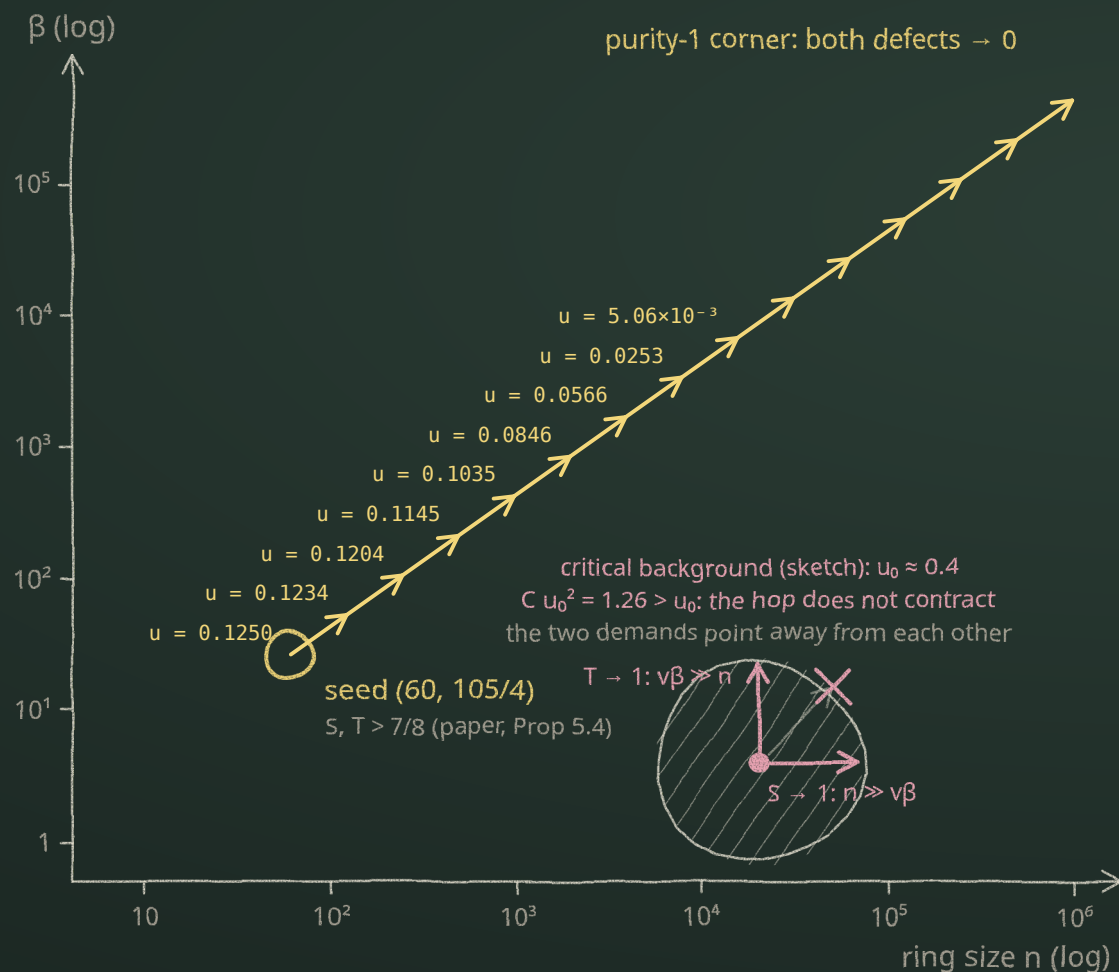
Square $p_k \propto e^{-\beta E_k}$ and renormalise: the Gibbs vector at 2β . Square $q_i \propto |\lambda_i|^n$: the spatial vector at $2n$. Lemma 4.1: defect $u \rightarrow$ at most $u^2/(1-u)^2$.

Live, Gibbs at $(4, 3)$: $u = 0.2607 \rightarrow$ at $(4, 6)$: 0.0148, bound 0.0621.

Spatial, three-state ring at $(8, 2)$: $u = 0.3482 \rightarrow$ at $(16, 2)$: 0.1427, bound 0.1427.

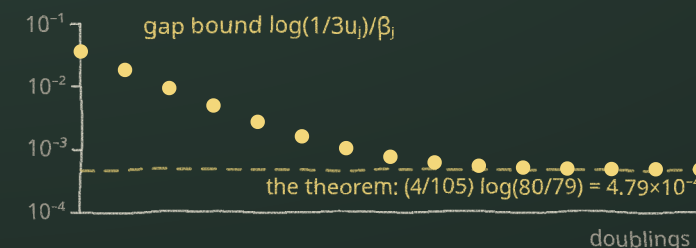
Squaring moves T along β and S along n . The identity trades one axis for the other.

The flow



$u_{j+1} = C u_j^2$, $\beta_j = 2^j \beta_0$, $n_j = 2^j n_0$, gap bound $\log(1/3u_j)/\beta_j$
 seed as in the paper: (60, 105/4), $u_0 = 1/8$, $C = 7.9$ (keys [] for u_0 , - = for C)

j	n_j	β_j	u_j	$\log(1/3u_j)/\beta_j$
0	60	26.25	0.1250	0.0374
1	120	52.50	0.1234	0.0189
2	240	105	0.1204	9.70×10^{-3}
3	480	210	0.1145	5.09×10^{-3}
4	960	420	0.1035	2.78×10^{-3}
5	1920	840	0.0846	1.63×10^{-3}
6	3840	1680	0.0566	1.06×10^{-3}
7	7680	3360	0.0253	7.67×10^{-4}
8	15360	6720	5.06×10^{-3}	6.23×10^{-4}
9	30720	13440	2.02×10^{-4}	5.51×10^{-4}
10	61440	26880	3.22×10^{-7}	5.15×10^{-4}
∞			$\rightarrow 0$	$\rightarrow \log(1/Cu_0)/\beta_0 = 4.79 \times 10^{-4}$



Gapless chain: finite-size gap $\approx v/n$, correlation length $\approx v\beta$. $T \rightarrow 1$ needs $v\beta \gg n$, $S \rightarrow 1$ needs $n \gg v\beta$. Best compromise: defects 0.3 to 0.5, above the 1/8 the machine needs. Gapped: ξ and $1/\Delta$ finite, the two demands decouple.

A finite ring that looks gapped on both axes with margin 1/8 looks gapped at every larger scale. Even sizes in between: Hölder on the spatial weights.

The fine print

what is proved, what was checked, where a bug would hide

- THEOREM 1.1** $H_L = \sum_j S_j S_{j+1}$, spin 1, periodic. Every even $L \geq 60$: unique ground state and $\gamma_L > (4/105) \log(80/79) \approx 4.8 \cdot 10^{-4}$. Every even $L \geq 2304$: $\gamma_L > \log 20 / 784 \approx 3.8 \cdot 10^{-3}$. Hence $\liminf \gamma_L > 0$.
- SCOPE** Even periodic rings and the $\liminf$ of their gaps. Cor 6.1 gives a gap inequality for subsequential limits of periodic ground states, **not** uniqueness or a gap for every infinite-volume ground state. True gap ≈ 0.41 : the constants are a thousand times smaller.
- THE SEEDS** Where the labour is. (60, 105/4) from exact Z_n , $n \leq 10$, at $\beta = 21/2$ (59,049 states, exact integers, rigorous series truncation), polynomial filters on the spectrum, a bond-dimension-26 MPS giving $Z_{120} > 1$, a 5/2-power raise of β . Margin: $S(60,105/4) > 7/8$ by $6 \cdot 10^{-4}$. Second seed (36, 49/4): defects 0.048 and 0.18, six doublings to (2304, 784).
- VERIFICATION** **No Lean**. Exact-rational scalar scripts and both MPS contractions re-run and pass; tightest margins $4.7 \cdot 10^{-5}$ and $6 \cdot 10^{-4}$. Independent exact diagonalisation reproduces the tables for $n \leq 9$. Unverified: $Z_{10}(21/2)$, which the $L \geq 60$ theorem needs, and $n = 10$ to 12 at $\beta = 49/4$.
- VERDICT** One reviewer, one pass: no mathematical error found. **85 to 90%** that the theorem is proved as written, conditional on the $n = 10$ to 12 integer totals. If a bug hides anywhere, it is in the seeds, not in the doubling.

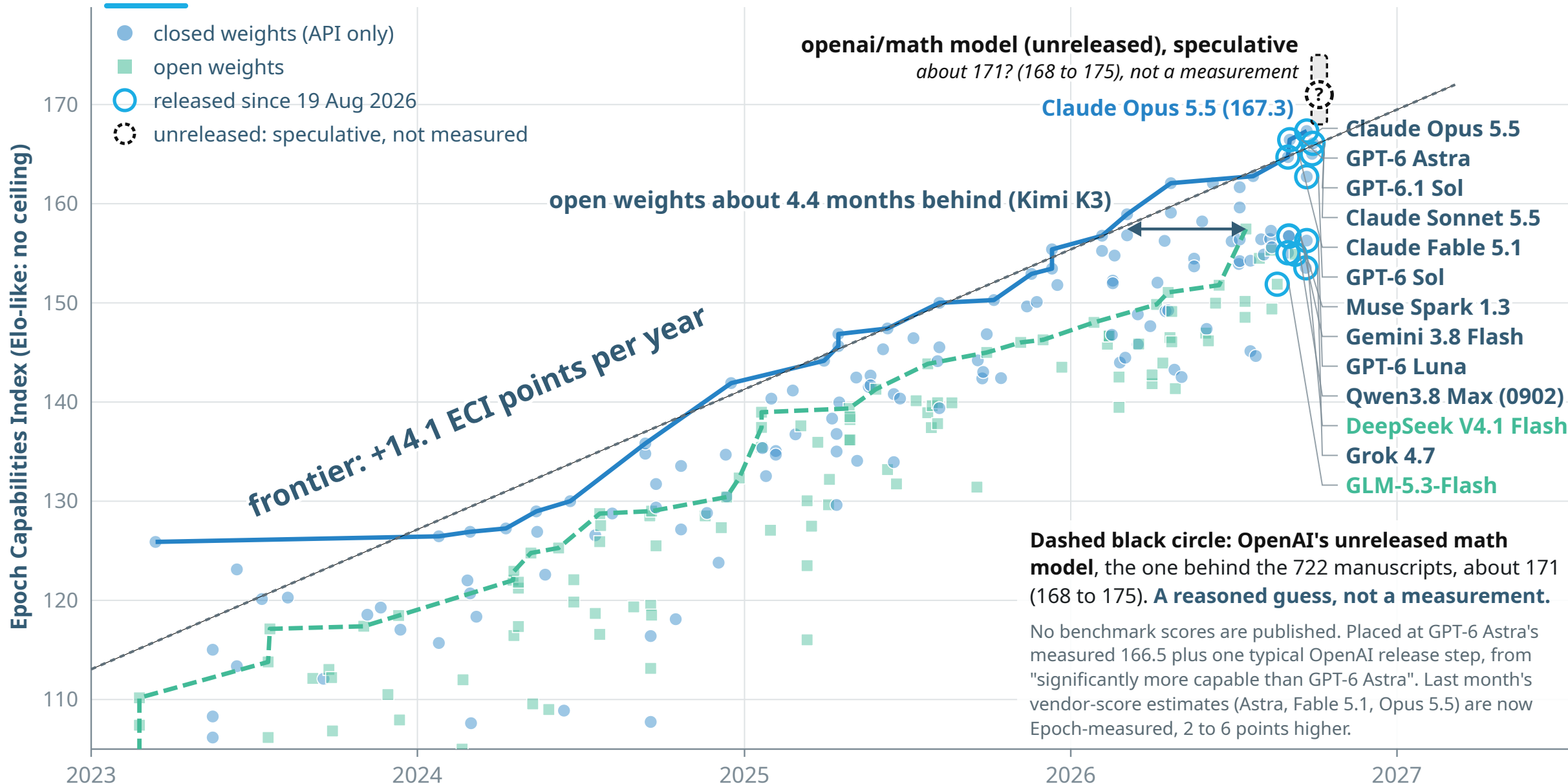
PART FIVE

The trend

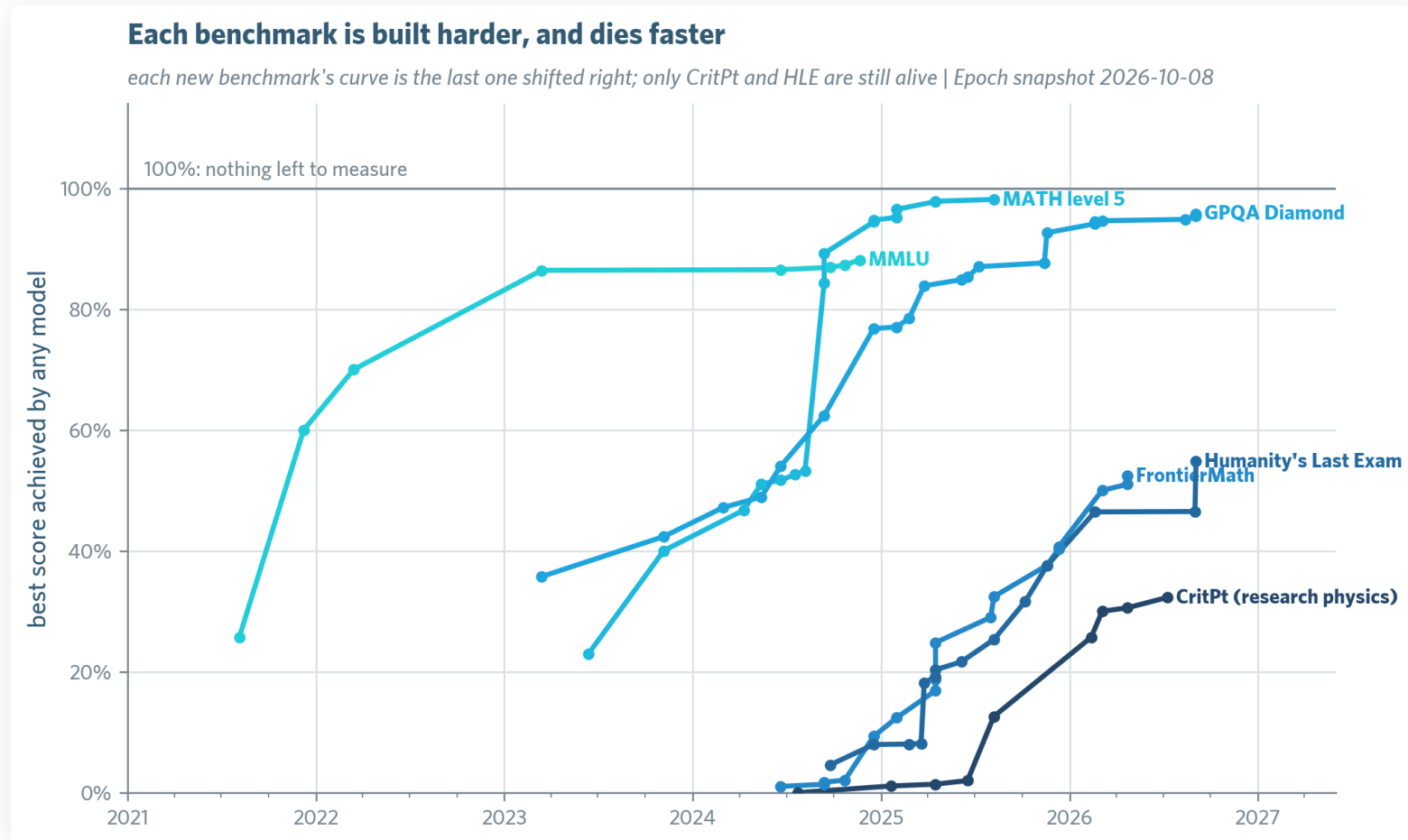
Capability is a straight line

all 272 measured models since 2023

Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-10-08



Benchmarks saturate, then get replaced



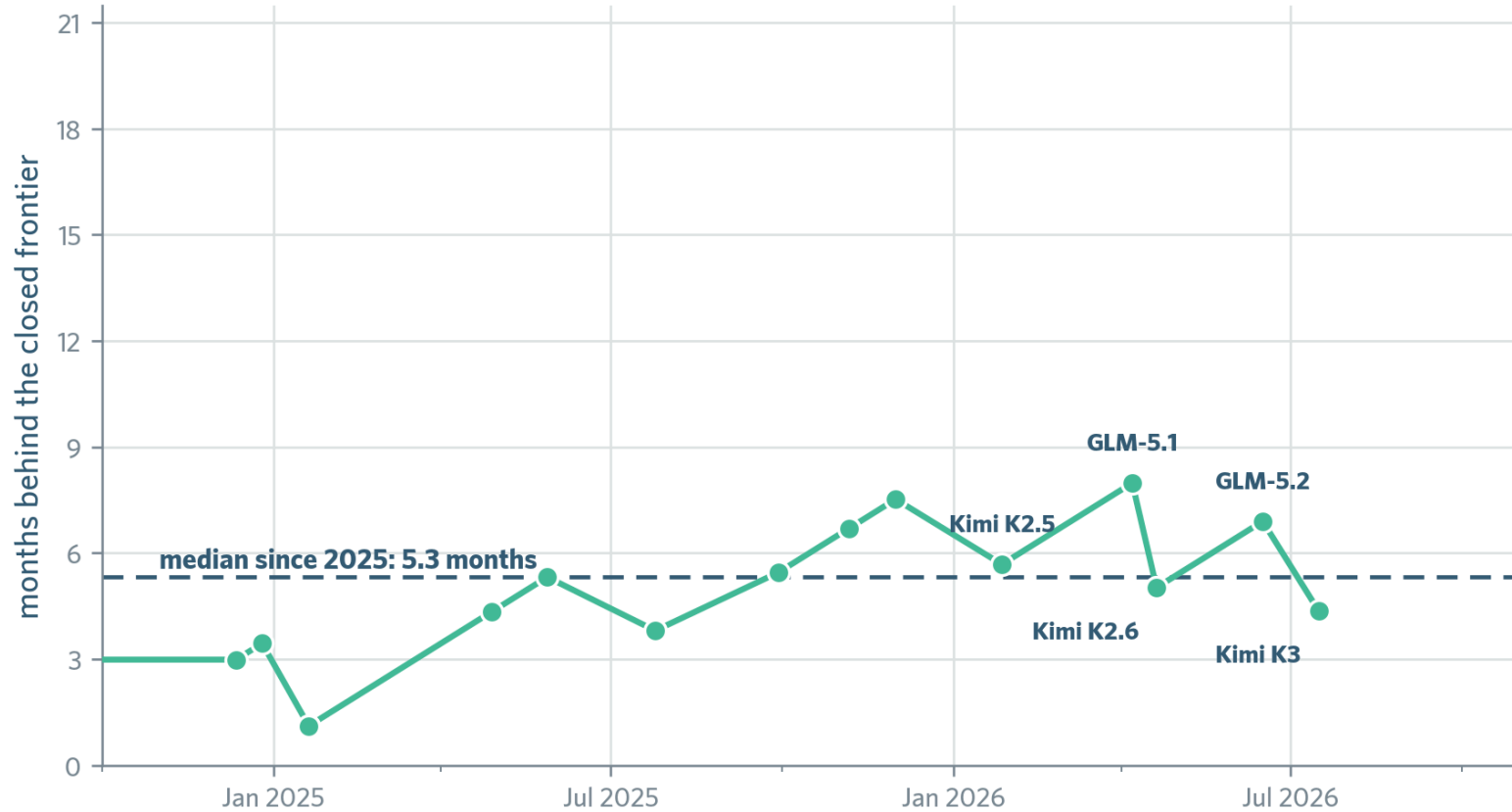
GPQA (PhD science QA) is **dead at 95.8%**; research physics (CritPt) still at **32%**; FrontierMath's original tiers are frozen at 52% because Epoch replaced them with harder v2 tiers

Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-10-08

You can run the frontier locally

Open weights trail the frontier by about 5 months

how long the closed frontier had already been at the level each new open model reaches | Epoch snapshot 2026-10-08



Qwen 3.8 27B: the first truly capable model that **fits one consumer GPU when quantised** (RTX 3090, 24 GB)
no vendor lock-in · Qwen 3.8 27B now Epoch-measured (ECI 149.4) · Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-10-08

**You can outsource cognition
but you can't outsource *understanding***

**The slop cannon is charged
whether we like it or not.
The only question is **how it is aimed****

Aimed with discipline (error-corrected, ground-truthed, adversarially checked),
it is the cheapest instrument for rigorous exploration the sciences have ever been handed.

How to train your slop cannon (Loader, Oppenheim, Osborne, 2026)
github.com/Open-Science-Ledger/how-to-train-your-slop-cannon